

TABLE 4.1 Mode frequencies and attenuations

Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}^a	q_{PREM}	ref. ^b	Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}	q_{PREM}	ref.
${}_0S_2$	309.25 ± 0.25	309.28	$2.30 \pm .25$	1.962	A	${}_0T_{15}$	2211.16 ± 0.40	2210.34	$6.45 \pm .45$	6.572	RR
${}_0T_2$	378.70 ± 0.50	379.17	$3.53 \pm .65$	3.994	A	${}_7S_1$		2224.25		4.018	
${}_2S_1$		403.96		2.520		${}_2S_9$	2229.30 ± 0.45	2228.75	$5.30 \pm .40$	5.312	ss
${}_0S_3$	468.46 ± 0.15	468.56	$2.46 \pm .15$	2.395	A	${}_2T_2$	2232.00 ± 1.00	2230.84	$4.04 \pm .40$	4.878	ss,RR
${}_0T_3$	585.45 ± 0.50	586.16	$4.45 \pm .50$	4.167	A	${}_0S_{14}$	2229.80 ± 0.25	2231.40	$3.15 \pm .10$	3.352	32^c ,RR
${}_0S_4$	646.82 ± 0.10	647.07	$2.75 \pm .25$	2.680	A	${}_3S_5$		2234.54		11.122	I
${}_1S_2$	680.13 ± 0.20	679.85	$3.12 \pm .20$	3.222	26^c ,A	${}_4S_4$	2278.50 ± 0.30	2279.60	$3.25 \pm .20$	3.446	26 ,ss
${}_0T_4$	765.64 ± 0.30	765.66	$3.92 \pm .30$	4.382	A	${}_1T_8$	2280.00 ± 0.70	2280.23	$3.81 \pm .35$	4.308	ss,RR
${}_0S_0$	814.38 ± 0.01	814.31	$0.19 \pm .03$	0.188	21^c ,A	${}_2T_3$	2294.90 ± 1.50	2294.97	$3.77 \pm .40$	4.836	ss
${}_0S_5$	840.01 ± 0.10	840.42	$2.92 \pm .15$	2.812	A	${}_0T_{16}$	2326.40 ± 0.30	2325.19	$6.76 \pm .50$	6.690	RR
${}_0T_5$	928.30 ± 0.25	928.24	$3.92 \pm .35$	4.621	RR,A	${}_0S_{15}$	2344.95 ± 0.15	2346.38	$3.18 \pm .10$	3.463	32^c ,RR
${}_2S_2$		937.85		10.432	I	${}_1S_{11}$	2346.20 ± 0.50	2347.54		2.674	W
${}_1S_3$	939.90 ± 0.20	939.83	$3.01 \pm .15$	3.537	27	${}_5S_4$	2379.25 ± 0.15	2379.52	$1.87 \pm .10$	2.044	27 ,ss
${}_3S_1$	944.35 ± 0.15	943.95	$1.17 \pm .05$	1.209	34^c	${}_2T_4$	2379.50 ± 0.80	2379.86	$4.11 \pm .40$	4.775	ss,RR
${}_0S_6$	1037.53 ± 0.10	1038.21	$3.04 \pm .10$	2.879	A	${}_2S_{10}$	2403.60 ± 0.50	2402.93	$5.50 \pm .30$	5.518	32 ,ss
${}_0T_6$	1079.20 ± 0.20	1078.83	$4.06 \pm .30$	4.868	RR,A	${}_6S_2$		2410.68		10.788	I
${}_3S_2$	1106.30 ± 0.30	1106.21	$3.30 \pm .30$	2.728	27^c	${}_4S_5$	2411.20 ± 0.30	2411.43	$3.45 \pm .20$	3.541	ss
${}_1S_4$	1172.78 ± 0.20	1172.85	$3.50 \pm .20$	3.689	27	${}_0T_{17}$	2440.32 ± 0.40	2439.09	$6.80 \pm .50$	6.795	RR
${}_0T_7$	1221.15 ± 0.20	1220.70	$4.80 \pm .30$	5.112	37 ,RR	${}_1T_9$	2452.20 ± 0.50	2452.49	$3.97 \pm .25$	4.399	RR
${}_0S_7$	1231.05 ± 0.10	1231.79	$2.95 \pm .15$	2.923	27	${}_0S_{16}$	2456.90 ± 0.15	2458.22	$3.25 \pm .10$	3.586	32^c ,RR
${}_1T_1$	1235.30 ± 0.50	1236.11	$3.50 \pm .30$	3.847	A,RR	${}_2T_5$	2484.90 ± 2.00	2485.09	$3.98 \pm .40$	4.693	ss
${}_2S_3$	1242.95 ± 0.15	1242.19	$2.50 \pm .20$	2.407	A	${}_2S_0$	2509.00 ± 0.35	2510.48	$0.83 \pm .05$	.805	21 ,A
${}_1T_2$	1319.15 ± 0.20	1320.13	$3.35 \pm .25$	3.901	A,RR	${}_7S_2$	2519.00 ± 1.50	2517.29	$3.55 \pm .50$	2.929	ss
${}_0T_8$	1356.60 ± 0.35	1356.11	$5.10 \pm .30$	5.346	34 ,RR	${}_3S_6$	2548.90 ± 0.35	2549.64	$3.25 \pm .35$	3.629	RR
${}_1S_5$	1370.00 ± 0.20	1370.27	$3.10 \pm .20$	3.426	27	${}_0T_{18}$	2553.35 ± 0.40	2552.22	$6.91 \pm .50$	6.890	32^c ,RR
${}_2S_4$	1379.70 ± 0.20	1379.20	$2.60 \pm .20$	2.630	27	${}_1S_{12}$		2555.06		2.738	M
${}_4S_1$	1411.70 ± 0.20	1412.64	$2.60 \pm .25$	2.817	ss	${}_0S_{17}$	2565.80 ± 0.30	2567.12	$3.28 \pm .11$	3.719	32^c ,RR
${}_0S_8$	1412.85 ± 0.10	1413.51	$2.92 \pm .10$	2.964	27^c ,RR	${}_2S_{11}$	2572.30 ± 0.35	2572.16		5.676	40 /RR
${}_3S_3$		1417.19		11.000	I	${}_2T_6$	2609.65 ± 2.00	2610.08	$4.16 \pm .50$	4.591	ss
${}_1T_3$	1438.50 ± 0.30	1439.13	$3.75 \pm .25$	3.955	A,RR	${}_1T_{10}$	2620.20 ± 0.80	2620.02	$3.93 \pm .35$	4.479	ss
${}_0T_9$	1487.07 ± 0.20	1486.61	$5.20 \pm .30$	5.566	RR	${}_4S_6$		2626.93		11.136	I
${}_2S_5$	1515.60 ± 0.30	1514.93	$3.30 \pm .20$	3.309	27	${}_0T_{19}$	2665.76 ± 0.20	2664.71	$7.04 \pm .55$	6.974	RR
${}_1S_6$	1521.50 ± 0.20	1522.04	$2.45 \pm .15$	2.893	27	${}_0S_{18}$	2672.50 ± 0.30	2673.30	$3.38 \pm .10$	3.860	RR
${}_0S_9$	1577.55 ± 0.15	1578.28	$2.90 \pm .15$	3.005	27^c ,RR	${}_3S_7$	2687.20 ± 1.20	2686.33	$3.52 \pm .30$	3.713	ss
${}_1T_4$	1585.50 ± 0.40	1585.50	$3.60 \pm .30$	4.006	A,RR	${}_5S_5$	2703.50 ± 0.20	2703.36	$1.83 \pm .10$	1.990	27 ,ss
${}_0T_{10}$	1613.67 ± 0.40	1613.26	$5.67 \pm .35$	5.773	RR	${}_2S_{12}$	2737.50 ± 0.60	2737.31	$4.58 \pm .50$	5.769	32
${}_1S_0$	1631.57 ± 0.05	1631.34	$0.50 \pm .03$	0.667	21^c ,A	${}_2T_7$	2751.50 ± 2.50	2753.73	$3.77 \pm .50$	4.476	ss
${}_1S_7$	1654.55 ± 0.15	1655.51	$2.40 \pm .15$	2.687	27	${}_1S_{13}$		2766.24		2.895	M
${}_2S_6$	1681.30 ± 0.30	1680.84	$4.10 \pm .20$	4.203	27	${}_0T_{20}$	2777.59 ± 0.40	2776.67	$7.19 \pm .60$	7.050	RR
${}_5S_1$		1713.79		11.019	I	${}_0S_{19}$	2776.10 ± 0.40	2776.98	$3.53 \pm .10$	4.006	RR
${}_4S_2$	1721.65 ± 0.20	1722.30	$2.05 \pm .15$	2.303	ss	${}_1T_{11}$	2783.00 ± 1.00	2783.31		4.546	WT
${}_0S_{10}$	1725.60 ± 0.20	1726.47	$2.92 \pm .10$	3.050	32^c ,RR	${}_3S_8$	2820.50 ± 0.60	2819.64	$3.65 \pm .25$	3.794	27
${}_0T_{11}$	1737.42 ± 0.20	1736.85	$5.85 \pm .35$	5.963	RR	${}_6S_3$	2821.85 ± 0.15	2821.72	$2.20 \pm .20$	2.345	27
${}_1T_5$	1750.15 ± 0.30	1750.49	$3.64 \pm .20$	4.061	A,RR	${}_8S_1$	2872.65 ± 0.20	2873.36	$0.95 \pm .10$	1.075	26 ,ss
${}_1S_8$	1797.80 ± 0.25	1799.30	$2.40 \pm .25$	2.635	27	${}_0S_{20}$	2877.95 ± 0.40	2878.37	$3.57 \pm .10$	4.154	33^c ,RR
${}_3S_4$		1833.30		11.113	I	${}_0T_{21}$	2888.67 ± 0.50	2888.20	$7.42 \pm .30$	7.118	RR,tvf
${}_0T_{12}$	1858.82 ± 0.20	1857.94	$5.95 \pm .35$	6.138	RR	${}_2S_{13}$	2897.00 ± 0.70	2899.90		5.738	W
${}_0S_{11}$	1861.30 ± 0.40	1862.42	$2.94 \pm .10$	3.105	RR	${}_2T_8$	2912.70 ± 1.50	2913.97	$3.58 \pm .80$	4.363	RR
${}_2S_7$	1865.20 ± 0.40	1864.96	$4.40 \pm .25$	4.726	ss	${}_1T_{12}$		2943.21		4.601	
${}_1T_6$	1925.30 ± 0.30	1925.61	$3.75 \pm .20$	4.129	ss,RR	${}_3S_9$	2951.00 ± 0.60	2951.58	$3.79 \pm .30$	3.865	34 ,RR
${}_1S_9$	1962.00 ± 0.40	1963.74	$2.45 \pm .25$	2.629	27	${}_1S_{14}$		2975.79		3.408	M
${}_0T_{13}$	1977.88 ± 0.20	1976.99	$6.17 \pm .45$	6.297	RR	${}_0S_{21}$	2977.30 ± 0.20	2977.73	$3.76 \pm .10$	4.303	33^c ,RR
${}_6S_1$	1983.20 ± 1.00	1980.38	$1.77 \pm .60$	1.538	ss	${}_0T_{22}$	3000.96 ± 0.50	2999.37	$7.60 \pm .30$	7.178	32^c ,G
${}_0S_{12}$	1989.00 ± 0.30	1990.37	$2.95 \pm .10$	3.173	32^c ,RR	${}_5S_6$	3011.55 ± 0.15	3010.69	$1.75 \pm .10$	1.975	27
${}_4S_3$	2048.35 ± 0.15	2048.97	$1.86 \pm .15$	2.083	27 ,ss	${}_4S_7$		3013.63		11.151	I
${}_2S_8$	2050.00 ± 0.35	2049.21	$5.10 \pm .30$	5.057	27	${}_2S_{14}$	3063.00 ± 2.00	3063.60		5.322	40 /34
${}_5S_2$	2090.60 ± 0.30	2091.28	$3.13 \pm .20$	3.146	ss	${}_0S_{22}$	3074.70 ± 0.20	3075.27	$4.02 \pm .10$	4.450	33^c ,tvf
${}_0T_{14}$	2095.43 ± 0.20	2094.36	$6.33 \pm .45$	6.442	RR	${}_7S_3$		3082.21		11.190	I
${}_1T_7$	2103.20 ± 0.30	2103.79	$3.60 \pm .30$	4.214	ss,RR	${}_3S_{10}$	3082.12 ± 0.50	3084.79		3.933	W
${}_0S_{13}$	2111.58 ± 0.10	2112.94	$3.11 \pm .10$	3.255	32^c ,RR	${}_2T_9$		3087.52		4.265	
${}_1S_{10}$	2146.50 ± 0.50	2148.42	$2.39 \pm .25$	2.643	WT	${}_6S_4$		3092.11		2.894	
${}_5S_3$	2168.80 ± 0.20	2169.66	$3.25 \pm .20$	3.420	27 ,ss	${}_1T_{13}$		3100.46		4.648	
${}_2T_1$	2187.00 ± 2.00	2187.83	$4.27 \pm .40$	4.904	ss	${}_0T_{23}$	3111.28 ± 0.50	3110.25	$7.30 \pm .40$	7.232	33^c ,tvf

TABLE 4.1 continued

Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}^a	q_{PREM}	ref. ^b	Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}	q_{PREM}	ref.
$1S_{15}$	3171.50 ± 1.50	3170.55		4.922	40/34	$3T_9$	3843.00 ± 5.00	3853.78		4.276	WT
$0S_{23}$	3170.70 ± 0.15	3171.26	$4.07 \pm .10$	4.594	$33^c, \text{tvf}$	$1T_{18}$		3859.82		4.874	
$9S_1$		3197.91		4.486		$4S_{10}$	3864.07 ± 0.70	3864.58		3.307	W
$3T_1$		3203.50		4.636		$2S_{18}$		3874.44		2.489	M
$8S_2$		3214.23		2.954		$9S_4$	3878.40 ± 1.50	3877.95	$1.71 \pm .20$	1.940	ss
$0T_{24}$	3222.50 ± 0.40	3220.90	$7.28 \pm .35$	7.281	$32^c, \text{tvf}$	$0T_{30}$	3882.90 ± 0.70	3881.75	$7.55 \pm .40$	7.479	$38, \text{tvf}$
$3S_{11}$	3219.52 ± 0.55	3221.27		4.002	W	$0S_{31}$	3904.70 ± 0.25	3905.40	$5.14 \pm .10$	5.571	$33, \text{tvf}$
$9S_2$	3230.50 ± 0.50	3231.75	$2.46 \pm .30$	2.456	ss	$1S_{20}$	3939.97 ± 0.80	3941.56		6.434	W
$3T_2$		3234.09		4.615		$7S_6$	3956.00 ± 0.50	3958.72	$1.85 \pm .20$	1.983	ss
$2S_{15}$		3240.89		3.878	M	$6S_9$	3964.50 ± 3.00	3965.35		3.119	WT
$1T_{14}$	3255.00 ± 1.50	3255.59		4.691	WT	$3S_{16}$	3966.85 ± 0.70	3966.87		4.408	W
$0S_{24}$	3265.50 ± 0.20	3265.89	$4.11 \pm .10$	4.734	$33^c, \text{tvf}$	$0T_{31}$	3992.60 ± 0.80	3991.62	$8.06 \pm .25$	7.501	G
$6S_5$		3266.82		3.683		$0S_{32}$	3994.00 ± 0.25	3995.04	$5.22 \pm .10$	5.671	$33, \text{H}, \text{tvf}$
$2T_{10}$	3267.50 ± 3.50	3270.21		4.193	WT	$3T_{10}$	3990.00 ± 5.00	3996.61		4.218	WT
$3S_0$	3272.52 ± 0.20	3271.18	$0.84 \pm .03$	.923	21,A	$1T_{19}$	4006.00 ± 2.00	4007.17		4.930	WT
$3T_3$		3279.74		4.585		$4S_{11}$	4007.10 ± 0.70	4010.47		3.513	W
$8S_3$		3283.64		3.916		$2T_{14}$	4011.00 ± 4.50	4016.39		4.155	WT
$5S_7$	3292.00 ± 0.30	3290.76	$1.90 \pm .10$	2.029	27	$10S_2$	4040.50 ± 1.00	4032.33	$1.17 \pm .15$	5.217	27
$0T_{25}$	3333.00 ± 0.60	3331.35	$7.32 \pm .30$	7.324	$32^c, \text{tvf}$	$11S_2$		4058.47		7.638	I
$1S_{16}$	3340.84 ± 1.50	3338.62		6.024	W	$0S_{33}$	4083.50 ± 0.30	4084.52	$5.35 \pm .10$	5.766	S, tvf
$3T_4$		3340.20		4.546		$1S_{21}$	4085.64 ± 0.65	4088.34		6.429	W
$0S_{25}$	3358.85 ± 0.15	3359.38	$4.29 \pm .10$	4.869	$33^c, \text{tvf}$	$2S_{19}$		4092.40		2.457	M
$3S_{12}$	3361.36 ± 0.45	3362.07		4.074	W	$0T_{32}$	4102.50 ± 0.80	4101.44	$7.65 \pm .40$	7.520	T, tvf
$10S_1$		3394.09		11.213	I	$4S_0$	4106.43 ± 0.25	4105.76	$0.79 \pm .03$	1.032	21,A
$4S_8$		3396.91		11.160	I	$3S_{17}$	4124.01 ± 0.60	4125.58		4.497	W
$6S_6$		3403.86		3.905		$3T_{11}$		4151.80		4.164	
$1T_{15}$	3407.00 ± 1.50	3408.95		4.733	WT	$4S_{12}$	4153.50 ± 1.00	4152.98		3.643	W
$7S_4$	3411.10 ± 0.60	3413.23		2.996		$1T_{20}$	4152.00 ± 2.50	4153.05		4.990	40/34
$3T_5$		3415.15		4.500		$5S_{10}$		4157.00		11.175	I
$0T_{26}$	3443.00 ± 0.60	3441.64	$7.41 \pm .30$	7.362	$32^c, \text{tvf}$	$8S_5$	4165.30 ± 0.25	4166.20	$1.60 \pm .15$	1.636	27
$2S_{16}$		3443.46		2.821	M	$0S_{34}$	4172.50 ± 0.30	4173.90	$5.48 \pm .10$	5.857	S, tvf
$0S_{26}$	3451.35 ± 0.15	3451.91	$4.44 \pm .10$	5.000	$33^c, \text{tvf}$	$2T_{15}$	4190.00 ± 2.00	4196.12		4.180	WT
$2T_{11}$	3456.00 ± 2.50	3457.67		4.150	WT	$6S_{10}$	4211.20 ± 0.30	4210.76		2.822	WT
$1S_{17}$	3493.39 ± 0.85	3493.94		6.309	W	$0T_{33}$	4212.20 ± 1.00	4211.23		7.537	T
$3T_6$		3504.29		4.449		$1S_{22}$	4231.47 ± 0.80	4234.40		6.418	W
$3S_{13}$	3507.55 ± 0.55	3507.44		4.152	W	$7S_7$	4234.10 ± 0.30	4237.85	$2.20 \pm .30$	2.409	ss
$5S_8$	3526.00 ± 0.60	3525.65	$2.30 \pm .15$	2.391	27	$0S_{35}$	4261.70 ± 0.30	4263.23	$5.57 \pm .10$	5.945	S, tvf
$0S_{27}$	3543.10 ± 0.15	3543.65	$4.49 \pm .10$	5.125	$33, \text{tvf}$	$3S_{18}$	4283.80 ± 0.70	4285.98		4.586	W
$0T_{27}$	3553.30 ± 0.50	3551.80	$7.35 \pm .30$	7.397	tvf	$4S_{13}$	4292.05 ± 1.10	4295.21		3.728	W
$6S_7$		3552.60		3.919		$1T_{21}$	4295.00 ± 1.00	4297.42		5.055	WT
$9S_3$	3556.50 ± 0.40	3554.98	$1.45 \pm .15$	1.286	27	$12S_1$		4300.34		4.466	
$1T_{16}$	3560.50 ± 1.00	3560.73		4.777	WT	$10S_3$		4300.92		3.451	
$3T_7$		3607.30		4.393		$4T_1$		4304.57		4.528	
$0S_{28}$	3634.20 ± 0.15	3634.76	$4.67 \pm .08$	5.244	$33, \text{tvf}$	$2S_{20}$		4310.64		2.440	M
$1S_{18}$	3643.75 ± 0.60	3644.94		6.395	W	$3T_{12}$	4317.00 ± 6.00	4318.29		4.120	WT
$2T_{12}$	3640.50 ± 2.50	3646.13		4.134	WT	$0T_{34}$	4322.20 ± 1.00	4321.00		7.553	T
$3S_{14}$	3656.20 ± 0.55	3657.11		4.234	W	$9S_5$		4324.87		11.117	I
$2S_{17}$		3657.42		2.564	M	$4T_2$		4326.83		4.527	
$7S_5$	3657.78 ± 0.35	3659.75	$2.05 \pm .20$	2.095	ss	$12S_2$		4330.16		4.357	
$0T_{28}$	3663.50 ± 0.50	3661.86	$7.35 \pm .30$	7.427	$32, \text{tvf}$	$0S_{36}$	4351.10 ± 0.30	4352.53	$5.66 \pm .10$	6.030	S, tvf
$11S_1$		3685.48		1.507		$4T_3$		4360.07		4.527	
$8S_4$		3702.73		10.692	I	$2T_{16}$	4367.00 ± 3.50	4372.15		4.211	WT
$4S_9$	3707.50 ± 1.50	3708.76		2.950	WT	$1S_{23}$	4376.25 ± 1.20	4379.84		6.405	W
$1T_{17}$	3710.00 ± 1.00	3711.01		4.824	WT	$10S_4$		4381.16		4.217	
$3T_8$		3723.89		4.335		$4T_4$		4404.14		4.527	
$0S_{29}$	3724.70 ± 0.20	3725.34	$4.76 \pm .10$	5.358	$33, \text{tvf}$	$0T_{35}$	4432.00 ± 1.00	4430.75		7.566	T
$6S_8$	3737.50 ± 2.00	3737.62		3.615	WT	$8S_6$		4435.24		2.268	
$0T_{29}$	3773.20 ± 0.70	3771.84	$7.45 \pm .40$	7.455	$32, \text{tvf}$	$4S_{14}$	4435.30 ± 0.85	4439.08		3.784	W
$5S_9$		3777.80		11.168	I	$1T_{22}$	4440.00 ± 2.00	4440.27		5.124	WT
$1S_{19}$	3792.23 ± 0.90	3793.89		6.426	W	$0S_{37}$	4440.20 ± 0.35	4441.84	$5.69 \pm .10$	6.111	S, H
$3S_{15}$	3810.98 ± 0.60	3810.48		4.319	W	$3S_{19}$	4446.13 ± 0.70	4447.56		4.675	W
$0S_{30}$	3814.80 ± 0.20	3815.52	$5.05 \pm .10$	5.467	$33, \text{tvf}$	$7S_8$	4448.00 ± 1.00	4452.59		3.111	WT
$2T_{13}$	3830.50 ± 3.50	3832.88		4.138	WT	$5S_{11}$	4458.00 ± 2.00	4456.55		2.669	

TABLE 4.1 continued

Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}^a	q_{PREM}	ref. ^b	Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}	q_{PREM}	ref.
$4T_5$		4458.81		4.526		$3T_{16}$	5040.00 ± 4.50	5054.35		4.121	WT
$11S_3$		4462.42		2.235		$0S_{44}$	5067.60 ± 0.60	5069.01		6.615	S
$10S_5$		4469.75		4.129		$7S_{12}$	5069.25 ± 1.60	5071.02		3.936	W
$3T_{13}$	4482.00 ± 7.00	4494.38		4.090	WT	$11S_5$	5072.60 ± 0.25	5074.41	$1.56 \pm .12$	1.503	27
$13S_1$	4494.60 ± 0.35	4495.73	$1.30 \pm .40$	1.360	WT	$0T_{41}$	5090.60 ± 1.50	5089.15		7.617	T
$4T_6$		4523.86		4.526		$1S_{28}$	5088.65 ± 2.60	5097.80		6.360	W
$1S_{24}$	4521.03 ± 1.20	4524.68		6.391	W	$3S_{23}$	5098.42 ± 0.80	5098.49		4.999	W
$2S_{21}$		4528.88		2.430	M	$4T_{12}$	5116.00 ± 6.00	5119.49		4.517	WT
$0S_{38}$	4529.50 ± 0.35	4531.20	$5.88 \pm .10$	6.190	S,tvf	$1T_{27}$	5129.00 ± 2.00	5131.50		5.525	WT
$6S_{11}$		4534.94		11.182	I	$5S_{14}$	5136.40 ± 0.80	5136.81		2.691	W
$0T_{36}$	4541.80 ± 1.00	4540.49		7.578	T	$9S_8$	5138.50 ± 3.50	5144.45		2.117	WT
$2T_{17}$	4543.00 ± 2.00	4544.86		4.245	WT	$0S_{45}$	5157.90 ± 0.60	5159.01		6.680	S
$1T_{23}$	4580.00 ± 2.00	4581.59		5.198	WT	$2S_{24}$		5182.42		2.447	M
$4S_{15}$	4584.00 ± 1.50	4585.74		3.820	W/WT	$13S_3$	5193.90 ± 0.25	5193.82	$1.15 \pm .10$	1.101	27
$4T_7$		4599.04		4.527		$0T_{42}$	5200.50 ± 1.50	5198.89		7.621	T
$3S_{20}$	4608.98 ± 0.90	4609.89		4.761	W	$4S_{19}$	5200.63 ± 1.55	5206.51		3.862	W
$7S_9$	4612.00 ± 2.50	4617.95		3.548	WT	$2T_{21}$	5202.00 ± 2.50	5210.67		4.399	WT
$0S_{39}$	4618.90 ± 0.35	4620.61	$5.97 \pm .10$	6.266	33,tvf	$8S_9$	5207.00 ± 2.00	5211.87		1.994	WT
$9S_6$		4620.88		3.020		$6S_{13}$	5233.50 ± 2.50	5233.88		3.944	WT
$0T_{37}$	4651.40 ± 1.00	4650.23		7.588	T	$1S_{29}$	5233.02 ± 1.50	5239.29		6.363	W
$8S_7$	4645.15 ± 0.60	4650.46		2.844		$3T_{17}$	5234.00 ± 8.00	5242.20		4.171	WT
$1S_{25}$	4662.65 ± 2.10	4668.93		6.378	W	$0S_{46}$	5248.20 ± 0.60	5249.12		6.743	S
$3T_{14}$		4677.69		4.078		$4T_{13}$	5243.00 ± 7.00	5251.78		4.505	WT
$12S_3$		4683.62		11.585	I	$3S_{24}$	5262.94 ± 0.85	5261.42		5.050	W
$4T_8$		4684.10		4.527		$1T_{28}$	5264.50 ± 2.00	5265.22		5.611	WT
$5S_{12}$	4696.80 ± 0.60	4695.98		2.593	W	$7S_{13}$		5288.15		11.194	I
$0S_{40}$	4708.40 ± 0.40	4710.10	$6.06 \pm .10$	6.339	S,tvf	$12S_4$		5293.80		11.333	I
$2T_{18}$	4710.00 ± 2.50	4714.71		4.281	WT	$15S_1$		5295.75		1.471	
$1T_{24}$	4721.00 ± 2.00	4721.36		5.276	WT	$0T_{43}$	5310.30 ± 1.50	5308.63		7.625	T
$4S_{16}$	4729.85 ± 1.50	4735.78		3.842	W	$5S_{15}$	5327.26 ± 1.20	5330.11		2.894	W
$2S_{22}$		4746.99		2.424	M	$0S_{47}$	5338.50 ± 0.60	5339.35		6.804	S
$0T_{38}$	4761.20 ± 1.30	4759.96		7.597	T	$11S_6$	5348.28 ± 0.50	5351.70	$2.31 \pm .25$	2.158	
$11S_4$	4766.05 ± 0.25	4766.86	$1.37 \pm .07$	1.425	27	$5T_1$		5353.50		5.012	
$7S_{10}$	4763.50 ± 4.50	4767.76		3.756	WT	$16S_1$		5355.04		4.984	
$3S_{21}$	4771.58 ± 0.80	4772.64		4.845	W	$4S_{20}$	5362.19 ± 1.30	5369.24		3.862	W
$4T_9$		4778.82		4.527		$5T_2$		5370.60		5.010	
$0S_{41}$	4798.10 ± 0.40	4799.67		6.411	S	$2T_{22}$	5366.00 ± 2.50	5372.32		4.440	WT
$1S_{26}$	4809.11 ± 1.40	4812.57		6.368	W	$14S_2$		5374.58		4.938	
$13S_2$	4844.70 ± 0.35	4845.26	$1.02 \pm .05$	1.138	27	$1S_{30}$	5373.06 ± 1.80	5379.96		6.371	W
$1T_{25}$	4859.50 ± 2.00	4859.60		5.357	WT	$9S_9$	5381.55 ± 3.00	5389.25		2.696	WT
$3T_{15}$		4865.33		4.089		$4T_{14}$	5384.50 ± 4.50	5393.54		4.488	WT
$0T_{39}$	4871.00 ± 1.30	4869.69		7.605	T	$5T_3$		5396.19		5.007	
$9S_7$		4872.64		2.041		$1T_{29}$	5399.50 ± 2.50	5397.51		5.697	WT
$2T_{19}$	4874.00 ± 3.00	4882.09		4.319	—	$2S_{25}$		5398.22		2.734	M
$4T_{10}$	4885.00 ± 5.00	4883.02		4.526	WT	$14S_3$		5407.52		4.866	
$5S_0$	4885.50 ± 0.30	4884.17	$0.94 \pm .03$	1.086	21,A	$6S_{14}$	5411.00 ± 2.00	5410.08		3.857	W/34
$0S_{42}$	4887.90 ± 0.50	4889.34		6.481	S	$0T_{44}$	5419.90 ± 1.50	5418.38		7.627	T
$4S_{17}$	4885.32 ± 1.20	4889.38		3.854	W	$3S_{25}$	5424.50 ± 2.50	5425.56		4.835	WT
$8S_8$	4902.34 ± 2.00	4908.07		2.384	WT	$3T_{18}$	5414.00 ± 7.00	5427.08		4.232	WT
$6S_{12}$		4911.92		11.188	I	$0S_{48}$	5429.00 ± 0.60	5429.70		6.865	S
$10S_6$		4914.11		11.470	I	$5T_4$		5430.22		5.002	
$7S_{11}$	4915.50 ± 3.00	4916.94		3.870	WT	$13S_4$		5455.12		4.811	
$5S_{13}$	4925.46 ± 1.10	4924.40		2.590	W	$5T_5$		5472.60		4.996	
$14S_1$		4925.38		11.350	I	$10S_7$		5489.22		11.287	I
$3S_{22}$	4932.87 ± 0.90	4935.56		4.926	W	$5S_{16}$	5505.95 ± 1.10	5506.96		3.117	W
$1S_{27}$	4952.77 ± 1.60	4955.54		6.361	W	$8S_{10}$	5506.78 ± 1.00	5508.75		2.031	W
$2S_{23}$		4964.89		2.425	M	$1S_{31}$	5513.54 ± 2.10	5519.76		6.385	W
$0S_{43}$	4977.70 ± 0.50	4979.12	$6.33 \pm .15$	6.549	S,H/G	$0S_{49}$	5519.50 ± 0.70	5520.16		6.924	S
$0T_{40}$	4980.70 ± 1.30	4979.42	$8.30 \pm .25$	7.611	T,G	$5T_6$		5523.25		4.987	
$1T_{26}$		4996.30		5.440		$12S_5$		5527.66		4.539	
$4T_{11}$	4993.50 ± 4.50	4996.59		4.523	WT	$0T_{45}$	5529.80 ± 1.50	5528.13		7.630	T
$4S_{18}$	5043.66 ± 1.10	5046.41		3.860	W	$1T_{30}$	5527.50 ± 1.50	5528.40		5.783	WT
$2T_{20}$	5044.50 ± 4.00	5047.32		4.359	WT	$2T_{23}$	5525.50 ± 2.50	5532.41		4.481	WT

TABLE 4.1 continued

Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}^a	q_{PREM}	ref. ^b	Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}	q_{PREM}	ref.
$4S_{21}$	5526.07 ± 1.30	5534.06	$1.30 \pm .15$	3.861	W	$7S_{15}$	6039.50 ± 3.50	6038.97	$1.85 \pm .10$	11.205	I
$14S_4$	5543.30 ± 0.60	5541.84		1.346	WT	$1T_{34}$		6039.22		6.113	WT
$4T_{15}$	5538.00 ± 7.00	5544.83		4.462	WT	$4T_{18}$		6051.83		4.362	
$15S_2$		5557.32		9.752	I	$3S_{28}$		6053.73		2.377	M
$11S_7$		5563.67		3.563		$11S_8$		6055.80		11.532	I
$5T_7$		5582.08		4.976		$0S_{55}$	6066.50 ± 1.50	6065.26		7.255	S
$2S_{26}$	5579.00 ± 3.00	5582.65		5.148	WT	$2S_{29}$	6068.14 ± 1.30	6066.18		5.460	W
$6S_{15}$	5600.20 ± 1.30	5602.51		3.673	W	$1S_{35}$	6061.83 ± 2.40	6069.14		6.494	W
$3T_{19}$	5607.00 ± 4.50	5608.11		4.296	WT	$0T_{50}$	6079.00 ± 3.00	6076.99		7.630	T
$0S_{50}$	5610.30 ± 0.70	5610.73		6.982	S	$5T_{13}$		6102.68		4.816	
$9S_{10}$	5605.54 ± 1.60	5610.93		3.127	W	$17S_{21}$	6122.60 ± 1.50	6129.05		1.396	
$3S_{26}$		5620.49		2.489	M	$3T_{22}$	6125.00 ± 6.00	6129.31		4.456	50
$0T_{46}$	5639.50 ± 2.00	5637.89		7.631	T	$12S_8$	6133.30 ± 0.40	6137.17		1.763	WT,ss
$12S_6$		5646.58		3.743		$5S_{20}$	6152.22 ± 1.90	6155.98		3.579	W
$5T_8$		5649.00		4.961		$0S_{56}$	6158.30 ± 1.50	6156.48		7.306	S
$1T_{31}$	5656.00 ± 3.00	5657.96		5.868	WT	$2T_{27}$	6151.00 ± 6.00	6159.71		4.633	WT
$1S_{32}$	5652.36 ± 2.40	5658.62		6.404	WT	$13S_6$		6161.19		1.541	
$7S_{14}$		5663.80		11.200	I	$1T_{35}$	6166.00 ± 3.50	6164.06		6.189	WT
$5S_{17}$	5669.85 ± 1.30	5673.69		3.295	W	$8S_{14}$	6165.07 ± 4.60	6170.65		3.904	WT
$2T_{24}$	5684.50 ± 4.00	5691.09		4.522	WT	$0T_{51}$	6188.80 ± 3.00	6186.78		7.628	T
$16S_2$		5697.12		3.042		$9S_{12}$	6185.07 ± 0.90	6187.26		2.150	W
$4S_{22}$	5695.00 ± 1.50	5700.49		3.861	W/WT	$10S_{10}$	6185.10 ± 1.40	6190.88		2.658	W
$0S_{51}$	5701.60 ± 0.80	5701.42	$7.30 \pm .15$	7.039	S,H	$1S_{36}$	6195.00 ± 3.00	6203.83		6.534	W/WT
$4T_{16}$	5709.00 ± 7.00	5705.48		4.431	WT	$4S_{25}$	6197.23 ± 1.60	6204.82		3.868	W
$8S_{11}$	5709.54 ± 1.50	5717.48		3.095	W	$5T_{14}$		6217.17		4.768	
$5T_9$		5723.92		4.943		$16S_3$		6222.36		11.598	I
$10S_8$	5735.00 ± 7.00	5737.09		3.550	WT	$2S_{30}$	6228.56 ± 1.20	6225.58		5.524	W
$6S_0$	5742.43 ± 0.30	5740.25	$0.94 \pm .03$	1.095	WT,A	$4T_{19}$		6234.50		4.334	
$2S_{27}$	5746.13 ± 0.90	5745.07		5.316	W	$6S_{18}$	6238.10 ± 1.40	6235.56		3.236	W
$0T_{47}$	5749.30 ± 2.00	5747.66		7.631	T	$0S_{57}$	6250.00 ± 1.30	6247.80		7.357	S
$3T_{20}$	5785.00 ± 8.00	5785.19		4.357	WT	$3S_{29}$		6270.74		2.372	M
$1T_{32}$	5786.50 ± 3.00	5786.24		5.952	WT	$1T_{36}$	6287.50 ± 3.00	6287.87		6.263	WT
$0S_{52}$	5792.50 ± 1.20	5792.22		7.095	S	$0T_{52}$	6298.50 ± 3.00	6296.58		7.626	T
$1S_{33}$	5788.81 ± 2.10	5796.49		6.429	W	$3T_{23}$	6299.00 ± 6.00	6297.60		4.491	WT
$5T_{10}$		5806.78		4.919		$2T_{28}$	6306.50 ± 5.50	6313.75		4.667	50
$6S_{16}$	5806.74 ± 1.40	5808.40		3.474	W	$5S_{21}$	6310.38 ± 2.20	6316.82		3.632	W
$13S_5$		5834.28		4.079		$8S_{15}$	6314.11 ± 6.50	6317.11		3.990	WT
$5S_{18}$	5829.20 ± 2.00	5835.59		3.422	W	$15S_4$	6323.30 ± 1.00	6332.35	$2.60 \pm .20$	2.507	
$3S_{27}$		5836.74		2.389	M	$1S_{37}$	6331.98 ± 3.30	6337.41		6.578	W
$2T_{25}$	5843.00 ± 3.50	5848.47		4.561	WT	$0S_{58}$	6341.80 ± 1.50	6339.21		7.406	S
$12S_7$		5855.88		2.360		$5T_{15}$		6339.76		4.713	
$0T_{48}$	5859.20 ± 2.00	5857.43		7.631	T	$4S_{26}$	6365.49 ± 1.30	6373.42		3.876	W
$4S_{23}$	5861.48 ± 1.40	5868.03		3.861	W	$2S_{31}$	6385.16 ± 1.20	6384.20		5.585	W
$4T_{17}$	5870.00 ± 9.00	5874.88		4.396	WT	$17S_2$		6395.21		4.327	
$8S_{12}$	5869.04 ± 2.50	5876.05		3.565	-	$6T_1$		6397.86		4.955	
$0S_{53}$	5883.80 ± 1.50	5883.13	$7.04 \pm .20$	7.149	S,O	$13S_7$	6393.00 ± 3.50	6398.97		2.613	
$9S_{11}$	5880.83 ± 0.90	5885.77		2.416	W	$18S_1$		6402.12		5.271	
$5T_{11}$		5897.53		4.891		$0T_{53}$	6408.50 ± 3.00	6406.38		7.623	T
$2S_{28}$	5903.81 ± 1.10	5906.03		5.393	W	$1T_{37}$	6412.50 ± 2.50	6410.74		6.333	WT
$1T_{33}$	5915.00 ± 3.50	5913.31		6.034	WT	$6T_2$		6412.65		4.948	
$14S_5$		5929.81		8.644	I	$7S_{16}$		6413.76		11.211	I
$1S_{34}$	5929.20 ± 2.90	5933.35		6.459	W	$4T_{20}$	6410.40 ± 5.00	6420.67		4.316	50
$10S_9$	5939.00 ± 4.50	5941.76		3.125	WT	$19S_1$		6427.44		11.165	I
$3T_{21}$	5955.00 ± 6.00	5958.71		4.411	WT	$0S_{59}$	6434.60 ± 1.70	6430.72		7.454	W
$0T_{49}$	5969.00 ± 2.00	5967.21	$8.37 \pm .25$	7.631	T,G	$6T_3$		6434.81		4.938	
$0S_{54}$	5975.10 ± 1.40	5974.14		7.203	S,G	$11S_9$	6435.50 ± 2.50	6437.12		1.594	WT
$5S_{19}$	5988.49 ± 2.20	5995.82		3.513	W	$6S_{19}$	6446.15 ± 1.80	6447.44		3.197	W
$5T_{12}$		5996.15		4.856		$17S_3$		6452.27		4.679	
$2T_{26}$	6001.00 ± 4.50	6004.65		4.598	WT	$10S_{11}$	6446.66 ± 2.50	6454.44		2.637	W
$6S_{17}$	6020.71 ± 1.20	6021.31		3.325	W	$3T_{24}$	6461.00 ± 7.00	6464.14		4.517	
$8S_{13}$	6018.93 ± 3.50	6024.43		3.778	WT	$6T_4$		6464.29		4.924	
$15S_3$	6030.90 ± 0.35	6035.22	$1.20 \pm .10$	1.241	WT	$8S_{16}$	6468.00 ± 6.50	6464.75		4.057	WT
$4S_{24}$	6028.67 ± 1.50	6036.26		3.864	W	$2T_{29}$	6459.50 ± 5.00	6466.83		4.698	50

TABLE 4.1 continued

Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}^a	q_{PREM}	ref. ^b	Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}	q_{PREM}	ref.
$1S_{38}$	6465.34 ± 2.70	6469.86		6.627	W	$10S_{13}$	6863.79 ± 2.90	6874.43		3.601	W
$5T_{16}$	6470.00 ± 5.00	6470.62		4.651		$4S_{29}$	6872.95 ± 1.30	6877.18		3.917	W
$15S_5$		6475.31		4.113		$0S_{64}$	6896.00 ± 5.00	6889.58		7.678	S
$5S_{22}$	6473.56 ± 1.90	6478.57		3.678	W	$18S_3$	6886.50 ± 0.80	6891.93	$1.23 \pm .08$	1.174	WT
$9S_{13}$	6479.69 ± 0.90	6483.50		2.063	W	$1T_{41}$	6893.00 ± 5.00	6894.00		6.580	50
$3S_{30}$		6487.70		2.368	M	$5T_{19}$	6901.00 ± 5.00	6914.69		4.433	
$14S_6$		6493.51		11.069	I	$2T_{32}$	6911.00 ± 4.50	6920.90		4.784	WT
$6T_5$		6501.07		4.907		$3S_{32}$		6921.36		2.363	M
$6T_{54}$	6518.50 ± 3.00	6516.18		7.620	T	$7S_{19}$	6919.81 ± 4.50	6921.84		4.175	W
$0S_{60}$	6526.90 ± 1.50	6522.32		7.501	W	$11S_{11}$	6915.00 ± 9.00	6922.23		3.088	WT
$1T_{38}$	6533.80 ± 4.00	6532.73		6.400	50	$20S_1$	6954.50 ± 1.50	6954.04	$1.45 \pm .25$	1.141	WT
$4S_{27}$	6535.54 ± 1.70	6541.80		3.886	W	$0T_{58}$	6959.20 ± 3.00	6955.42	$8.50 \pm .30$	7.604	T,G
$2S_{32}$	6541.09 ± 1.50	6542.03		5.643	W	$6T_{12}$		6956.77		4.714	
$6T_6$		6545.07		4.887		$3T_{27}$	6949.50 ± 5.00	6956.93		4.557	50
$18S_2$	6537.00 ± 2.00	6545.69	1.79 ± 0.30	1.876	ss	$5S_{25}$	6965.94 ± 2.00	6967.92		3.803	W
$13S_8$	6552.00 ± 3.50	6553.59		3.591		$4T_{23}$	6978.00 ± 7.00	6980.21		4.309	50
$16S_4$		6568.64		3.389		$0S_{65}$	6989.00 ± 5.00	6981.61	$8.15 \pm .25$	7.720	S,G
$7S_0$	6584.80 ± 0.70	6580.71	1.12 ± 0.13	1.135		$1S_{42}$		6988.22		6.852	
$15S_6$		6595.93		4.489		$2S_{35}$	7011.91 ± 2.30	7010.52		5.800	W
$6T_7$		6596.24		4.864		$1T_{42}$	7013.50 ± 3.50	7013.05		6.633	WT
$1S_{39}$	6594.33 ± 3.40	6601.16		6.679	W	$9S_{15}$	7026.49 ± 1.20	7029.82		2.325	WT
$4T_{21}$	6600.00 ± 6.00	6608.12		4.307	50	$10S_{14}$	7031.00 ± 4.00	7038.76		3.859	WT
$5T_{17}$	6596.00 ± 7.00	6609.94		4.583		$4S_{30}$	7038.11 ± 1.30	7043.91		3.938	W
$12S_9$		6610.23		11.602	I	$14S_8$	7039.00 ± 1.50	7047.89		2.070	WT
$0S_{61}$	6619.00 ± 3.00	6614.00		7.547	S	$6S_{22}$	7050.31 ± 1.70	7048.18		3.324	W
$7S_{17}$	6610.15 ± 3.80	6614.22		4.110	W	$6T_{13}$		7049.19		4.680	T
$2T_{30}$	6609.00 ± 3.50	6619.02		4.728	WT	$0T_{59}$	7069.30 ± 3.00	7065.24		7.599	
$0T_{55}$	6628.50 ± 3.00	6625.98		7.617	T	$2T_{33}$	7064.00 ± 4.00	7070.65		4.811	50
$3T_{25}$	6622.50 ± 3.50	6629.35		4.536	WT	$0S_{66}$	7081.00 ± 5.00	7073.71		7.760	S
$5S_{23}$	6635.43 ± 1.80	6641.17		3.720	W	$19S_2$		7078.81		11.007	I
$1T_{39}$	6655.00 ± 4.00	6653.89		6.464	WT	$5T_{20}$	7074.00 ± 5.00	7080.11		4.356	
$6S_{20}$	6653.89 ± 1.40	6654.48		3.202	W	$7S_{20}$	7077.02 ± 3.60	7081.96		4.182	W
$6T_8$		6654.50		4.838		$15S_7$		7085.82		11.488	I
$13S_9$		6686.16		3.861		$1S_{43}$		7115.01		6.914	
$10S_{12}$	6687.00 ± 3.00	6687.11		3.104	W	$3T_{28}$	7110.00 ± 5.00	7119.64		4.562	50
$2S_{33}$	6697.19 ± 1.40	6699.04		5.698	W	$5S_{26}$	7132.66 ± 1.80	7131.42		3.846	W
$3S_{31}$		6704.57		2.365	M	$1T_{43}$	7131.00 ± 5.00	7131.51		6.683	50
$0S_{62}$	6711.00 ± 3.00	6705.78	$8.23 \pm .25$	7.592	S,H	$3S_{33}$		7138.07		2.360	M
$4S_{28}$	6702.65 ± 1.90	6709.77		3.900	W	$12S_{11}$	7135.66 ± 0.90	7138.82		1.956	W
$11S_{10}$	6706.02 ± 1.30	6712.42		2.345	W	$6T_{14}$		7148.14		4.644	
$6T_9$		6719.78		4.810		$11S_{12}$	7144.00 ± 5.00	7149.61		2.725	WT
$1S_{40}$	6728.29 ± 3.90	6731.31		6.734	W	$16S_6$	7149.30 ± 0.40	7153.68	$1.75 \pm .15$	1.352	WT
$0T_{56}$	6738.80 ± 3.00	6735.79		7.613	T	$13S_{10}$		7155.54		11.652	I
$5T_{18}$	6759.00 ± 5.00	6757.93		4.509		$8S_{18}$		7162.43		11.222	I
$7S_{18}$	6762.70 ± 6.00	6766.28		4.150	WT	$4T_{24}$	7156.00 ± 8.00	7163.04		4.313	50
$9S_{14}$	6766.50 ± 1.50	6768.24		2.063	W	$2S_{36}$	7164.41 ± 2.50	7164.96		5.847	W
$2T_{31}$	6764.00 ± 4.50	6770.36		4.756	50	$0S_{67}$	7175.00 ± 6.00	7165.89		7.799	S
$14S_7$	6773.00 ± 2.50	6772.89		3.026		$0T_{60}$	7179.20 ± 3.00	7175.06		7.594	T
$1T_{40}$	6773.50 ± 4.00	6774.29		6.524	WT	$10S_{15}$	7196.00 ± 3.00	7209.25		3.746	W
$8S_{17}$		6788.22		11.217	I	$4S_{31}$	7204.41 ± 2.20	7209.87		3.963	W
$6T_{10}$		6791.98		4.780		$2T_{34}$	7213.00 ± 4.00	7219.63		4.838	50
$3T_{26}$	6792.00 ± 5.00	6793.54		4.548	50	$6S_{23}$	7234.75 ± 1.50	7233.92		3.425	W
$4T_{22}$	6787.00 ± 5.00	6795.04		4.306	50	$9S_{16}$	7232.73 ± 2.60	7239.58		3.133	W
$0S_{63}$	6803.00 ± 5.00	6797.64	$8.04 \pm .20$	7.635	S,O	$1S_{44}$		7240.71		6.977	
$5S_{24}$	6800.78 ± 2.40	6804.39		3.761	W	$18S_4$	7238.30 ± 0.30	7241.00	$1.09 \pm .05$	1.060	WT
$16S_5$	6830.00 ± 0.50	6836.41	$1.85 \pm .10$	1.720	WT	$7S_{21}$	7248.37 ± 2.90	7247.61		4.166	W
$0T_{57}$	6849.00 ± 3.00	6845.61		7.609	T	$1T_{44}$	7248.50 ± 3.00	7249.42		6.730	50
$17S_4$		6854.04		11.690	I	$6T_{15}$		7253.56		4.609	
$6S_{21}$	6855.20 ± 1.80	6855.03		3.246	W	$5T_{21}$		7253.69		4.286	
$2S_{34}$	6852.39 ± 1.50	6855.21		5.750	W	$0S_{68}$	7268.00 ± 6.00	7258.14		7.838	S
$1S_{41}$		6860.33		6.792		$3T_{29}$	7271.00 ± 4.50	7281.77		4.565	50
$12S_{10}$	6860.00 ± 5.00	6861.33		3.022	WT	$0T_{61}$	7290.00 ± 3.00	7284.88		7.588	T
$6T_{11}$		6871.01		4.748		$5S_{27}$	7291.93 ± 2.30	7294.59		3.892	W

TABLE 4.1 continued

Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}^a	q_{PREM}	ref. ^b	Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}	q_{PREM}	ref.
$2S_{37}$	7318.55 ± 2.10	7318.53		5.890	W	$2T_{37}$	7653.00 ± 3.00	7661.93		4.924	WT
$4T_{25}$	7336.00 ± 9.00	7343.40		4.318	50	$15S_8$		7664.98		11.231	I
$0S_{69}$	7361.00 ± 7.00	7350.47		7.875	S	$7T_7$		7667.63		4.741	
$14S_9$	7345.00 ± 0.80	7354.08	$2.13 \pm .15$	1.892	WT	$10S_{17}$	7673.49 ± 1.40	7675.89		2.580	W
$3S_{34}$		7354.71		2.358	M	$20S_3$		7685.74		7.229	I
$20S_2$	7360.00 ± 3.50	7357.09	$1.73 \pm .35$	1.940	ss	$11S_{14}$	7679.00 ± 3.50	7686.80		2.504	WT
$1S_{45}$		7365.35		7.041		$16S_8$		7689.68		4.011	
$6T_{16}$		7365.37		4.574		$13S_{11}$		7693.02		11.669	I
$1T_{45}$	7362.50 ± 4.50	7366.83		6.773	WT	$4T_{27}$	7694.00 ± 7.00	7697.46		4.326	WT
$2T_{35}$	7362.00 ± 4.00	7367.85		4.865	WT	$9S_{19}$	7689.69 ± 4.50	7698.29		4.030	WT
$4S_{32}$	7369.39 ± 1.60	7374.98		3.992	W	$4S_{34}$	7700.08 ± 1.90	7702.48		4.062	W
$0T_{62}$	7290.00 ± 3.00	7394.70		7.582	T	$1T_{48}$	7715.50 ± 5.50	7716.44		6.886	50
$9S_{17}$	7398.33 ± 3.00	7403.56		3.675	WT	$7T_8$		7717.14		4.735	
$6S_{24}$	7412.49 ± 2.10	7413.13		3.534	W	$0S_{73}$	7736.00 ± 7.00	7720.47		8.012	S
$11S_{13}$	7411.50 ± 4.00	7417.49		2.379	WT	$0T_{65}$	7731.00 ± 3.00	7724.15		7.563	T
$7S_{22}$	7418.73 ± 2.00	7419.40		4.127	W	$1S_{48}$		7733.33		7.236	
$10S_{16}$	7420.12 ± 1.60	7422.64		3.038	W	$6T_{19}$		7738.59		4.474	
$8S_0$	7429.00 ± 1.00	7424.13	$0.98 \pm .12$	1.174	WT	$6S_{26}$	7756.13 ± 2.10	7757.89		3.733	W
$5T_{22}$		7434.43		4.226		$3T_{32}$	7757.50 ± 4.50	7764.94		4.572	WT
$0S_{70}$	7455.00 ± 7.00	7442.86		7.911	S	$15S_9$	7768.00 ± 3.50	7771.75		4.008	
$3T_{30}$	7439.00 ± 4.50	7443.36		4.567	WT	$7T_9$		7772.53		4.730	
$12S_{12}$	7449.50 ± 1.00	7455.08		1.754	W	$2S_{40}$	7774.94 ± 2.50	7773.95		6.005	W
$5S_{28}$	7455.36 ± 2.10	7457.18		3.939	W	$12S_{13}$	7767.92 ± 0.70	7777.00		1.758	W
$17S_5$		7461.22		10.212	I	$5S_{30}$	7778.52 ± 2.60	7780.00		4.034	W
$2S_{38}$	7473.00 ± 1.80	7471.21		5.931	W	$7S_{24}$	7778.87 ± 2.30	7780.08		4.013	W
$16S_7$	7471.50 ± 1.00	7474.14	$1.45 \pm .12$	1.250	WT	$3S_{36}$		7787.76		2.355	M
$6T_{17}$		7483.49		4.540		$21S_3$		7801.59		5.862	
$1T_{46}$	7483.50 ± 4.00	7483.78		6.813	WT	$17S_8$		7805.05		1.838	
$1S_{46}$		7488.98		7.106		$2T_{38}$	7798.00 ± 4.00	7807.76		4.956	50
$21S_1$		7495.27		4.718		$5T_{24}$		7810.82		4.155	
$7T_1$		7498.57		4.762		$0S_{74}$	7830.00 ± 7.00	7813.15	$8.73 \pm .25$	8.043	S,O
$0T_{63}$	7511.00 ± 3.00	7504.52		7.576	T	$22S_1$	7822.40 ± 0.50	7819.54	$0.95 \pm .05$	1.304	WT
$19S_3$		7504.81		4.384		$14S_{11}$	7813.16 ± 7.00	7823.54		3.769	WT
$7T_2$		7511.20		4.760		$1T_{49}$	7830.00 ± 6.00	7832.22		6.918	WT
$21S_2$		7514.86		4.577		$7T_{10}$		7833.70		4.725	
$2T_{36}$	7505.00 ± 5.00	7515.28		4.894	WT	$0T_{66}$	7842.00 ± 3.00	7833.97		7.555	T
$18S_5$		7517.39		4.365		$8S_{20}$	7846.85 ± 6.50	7843.35		4.106	WT
$4T_{26}$	7512.00 ± 5.00	7521.46		4.322	WT	$1S_{49}$		7854.14		7.300	
$7T_3$		7530.11		4.758		$4S_{35}$	7859.58 ± 1.80	7864.82		4.102	W
$0S_{71}$	7548.00 ± 7.00	7535.33		7.946	S	$4T_{28}$	7863.00 ± 8.00	7871.66		4.331	WT
$8S_{19}$		7536.41		11.229	I	$6T_{20}$		7875.57		4.443	
$4S_{33}$	7536.51 ± 1.90	7539.19		4.025	W	$15S_{10}$	7896.00 ± 6.00	7897.49		3.224	WT
$19S_4$		7540.90		4.538		$23S_1$		7899.09		11.631	I
$9S_{18}$	7541.47 ± 4.00	7552.78		3.907	W	$7T_{11}$		7900.55		4.720	
$7T_4$		7555.27		4.754		$0S_{75}$	7925.00 ± 7.00	7905.88		8.074	S
$3S_{35}$		7571.27		2.357	M	$9S_{20}$		7910.21		11.235	I
$17S_6$		7580.67		4.230		$11S_{15}$		7919.06		3.012	
$7T_5$		7586.61		4.750		$2S_{41}$	7921.24 ± 3.90	7924.01		6.038	W
$6S_{25}$	7588.13 ± 2.10	7587.27		3.640	W	$3T_{33}$	7914.50 ± 4.50	7924.89		4.576	WT
$7S_{23}$	7593.92 ± 1.70	7597.20		4.073	W	$6S_{27}$	7921.94 ± 1.70	7926.36		3.810	W
$1T_{47}$	7597.00 ± 5.00	7600.31		6.851	50	$10S_{18}$	7936.46 ± 1.10	7938.48		2.434	W
$3T_{31}$	7600.00 ± 5.00	7604.42		4.569	50	$5S_{31}$	7941.78 ± 2.40	7940.16		4.080	W
$6T_{18}$		7607.90		4.506		$0T_{67}$	7952.00 ± 4.00	7943.78	$8.46 \pm .30$	7.548	T,G
$1S_{47}$		7611.63		7.171		$1T_{50}$	7948.50 ± 3.50	7947.68		6.947	WT
$0T_{64}$	7621.00 ± 3.00	7614.33		7.569	T	$2T_{39}$	7943.50 ± 3.00	7952.75		4.990	WT
$5S_{29}$	7616.88 ± 2.50	7619.01		3.987	W	$13S_{12}$	7950.00 ± 5.50	7955.03		4.043	WT
$5T_{23}$		7620.80		4.181		$18S_6$		7957.04		2.530	
$2S_{39}$	7623.39 ± 2.70	7623.02		5.970	W	$7S_{25}$	7964.31 ± 2.80	7966.58		3.960	W
$7T_6$		7624.09		4.746		$7T_{12}$		7972.96		4.715	
$0S_{72}$	7642.00 ± 7.00	7627.87		7.979	S	$1S_{50}$		7974.10		7.363	
$17S_7$		7635.12		4.372		$8S_{21}$	7976.33 ± 4.25	7989.31		4.156	
$14S_{10}$	7624.00 ± 4.00	7635.75		2.574	WT	$0S_{76}$	8019.00 ± 7.00	7998.68	$8.90 \pm .25$	8.103	S,G/H
$19S_5$		7661.84		2.726		$5T_{25}$		8002.38		4.148	

TABLE 4.1 continued

Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}^a	q_{PREM}	ref. ^b	Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}	q_{PREM}	ref.
$3S_{37}$		8004.19		2.353	M	$4T_{31}$		8385.31		4.361	
$6T_{21}$		8018.91		4.413		$3T_{36}$	8395.00 ± 10.0	8401.09		4.598	WT
$4S_{36}$	8019.95 ± 1.50	8026.22		4.146	W	$1T_{54}$	8402.00 ± 6.00	8406.75		7.044	50
$4T_{29}$		8044.29		4.339		$12S_{15}$	8402.99 ± 1.00	8411.28		1.840	W
$7T_{13}$		8050.83		4.711		$7T_{17}$		8414.71		4.696	
$0T_{68}$	8062.00 ± 4.00	8053.59		7.540	T	$5S_{34}$	8409.00 ± 4.00	8416.62		4.191	W/WT
$1T_{51}$	8058.00 ± 6.00	8062.83		6.975	50	$11S_{18}$	8420.49 ± 9.00	8418.49		3.976	WT
$2S_{42}$	8072.00 ± 4.80	8073.21		6.069	W	$6S_{30}$	8417.21 ± 3.10	8427.41		3.952	W
$3T_{34}$	8080.00 ± 5.00	8084.26		4.581	WT	$15S_{12}$	8426.74 ± 2.00	8432.73		1.747	W
$13S_{13}$	8084.46 ± 9.00	8091.46		3.907		$3S_{39}$		8436.86		2.350	M
$0S_{77}$	8112.00 ± 7.00	8091.55		8.131	S	$16S_{10}$	8433.00 ± 3.00	8437.72		1.291	WT
$19S_6$		8093.14		10.515	I	$8S_{24}$	8434.49 ± 7.00	8440.00		4.211	—
$1S_{51}$		8093.25		7.425		$1S_{54}$		8446.31		7.603	
$6S_{28}$	8088.18 ± 2.10	8093.69		3.871	W	$10S_{20}$	8444.79 ± 1.40	8446.66		2.560	W
$2T_{40}$	8089.00 ± 5.00	8096.87		5.027	WT	$0S_{81}$	8490.00 ± 8.00	8463.61		8.232	S
$12S_{14}$	8088.30 ± 0.90	8097.37		1.809	W	$20S_5$	8465.50 ± 2.00	8471.58	$1.76 \pm .25$	1.571	WT
$5S_{32}$	8099.06 ± 2.10	8099.56		4.122	W	$13S_{15}$	8469.42 ± 9.00	8474.42		2.968	WT
$11S_{16}$	8101.00 ± 5.00	8107.03		3.514	WT	$6T_{24}$		8487.69		4.332	
$20S_4$	8118.00 ± 2.00	8116.77	$1.38 \pm .25$	1.278	ss	$0T_{72}$	8503.00 ± 4.00	8492.80		7.507	T
$16S_9$	8116.00 ± 2.00	8117.59		1.392	WT	$4S_{39}$	8499.51 ± 2.40	8504.92		4.293	W
$15S_{11}$	8122.42 ± 2.00	8130.95		1.999	W	$2S_{45}$	8508.96 ± 4.50	8515.75		6.150	W
$7T_{14}$		8134.04		4.707		$7T_{18}$		8518.35		4.691	
$8S_{22}$	8127.86 ± 4.40	8136.99		4.188	WT	$1T_{55}$	8514.40 ± 7.00	8520.94		7.063	50
$7S_{26}$	8154.33 ± 2.30	8155.16		3.919	W	$2T_{43}$	8507.00 ± 8.00	8523.66		5.153	WT
$0T_{69}$	8173.00 ± 4.00	8163.40		7.533	T	$7S_{28}$	8522.63 ± 3.00	8533.66		3.876	W
$6T_{22}$		8168.68		4.385		$24S_1$		8550.62		4.963	
$1T_{32}$	8171.00 ± 7.00	8177.72		7.000	50	$8T_1$		8551.15		4.975	
$0S_{78}$	8207.00 ± 8.00	8184.47		8.158	S	$4T_{32}$	8548.00 ± 7.00	8553.86		4.377	WT
$4S_{37}$	8184.38 ± 1.50	8186.69		4.192	W	$0S_{82}$	8585.00 ± 8.00	8556.77		8.254	S
$5T_{26}$		8193.48		4.159		$3T_{37}$	8548.00 ± 5.00	8558.49		4.609	WT
$10S_{19}$	8195.52 ± 1.10	8197.96		2.428	W	$22S_3$		8559.91		3.686	
$22S_2$	8200.00 ± 2.00	8207.04	$1.67 \pm .25$	1.397	41	$23S_2$		8561.36		5.004	
$1S_{52}$		8211.63		7.486		$8T_2$		8561.68		4.974	
$4T_{30}$	8196.00 ± 10.0	8215.47		4.348	50	$1S_{55}$		8562.68		7.659	
$3S_{38}$		8220.55		2.352	M	$11S_{19}$		8568.59		4.020	
$2S_{43}$	8219.54 ± 4.10	8221.56		6.098	W	$5T_{28}$		8568.60		4.218	
$7T_{15}$		8222.48		4.703		$5S_{35}$	8570.59 ± 2.90	8574.62		4.218	W
$14S_{12}$		8224.06		11.684	I	$8T_3$		8577.46		4.973	
$2T_{41}$	8230.00 ± 8.00	8240.08		5.066	WT	$24S_2$		8585.28		11.454	I
$3T_{35}$	8234.00 ± 10.0	8243.00		4.589	50	$6S_{31}$	8588.30 ± 2.70	8594.46		3.975	W
$17S_9$		8248.53		11.488	I	$8S_{25}$	8590.61 ± 7.50	8596.24		4.205	—
$5S_{33}$	8253.59 ± 2.60	8258.32		4.159	W	$22S_4$		8598.40		4.877	
$13S_{14}$	8254.84 ± 6.50	8258.94		3.502	WT	$8T_4$		8598.47		4.970	
$6S_{29}$	8255.78 ± 2.10	8260.57		3.918	W	$18S_8$		8599.27		2.423	
$9S_0$	8268.00 ± 2.00	8262.64	$0.80 \pm .15$	1.190	WT	$0T_{73}$	8613.00 ± 5.00	8602.59		7.499	T
$11S_{17}$	8265.00 ± 8.00	8268.00		3.821	WT	$23S_3$		8602.59		2.605	
$0T_{70}$	8282.00 ± 4.00	8273.20		7.524	T	$8T_5$		8624.69		4.967	
$0S_{79}$	8302.00 ± 8.00	8277.46		8.184	S	$7T_{19}$		8626.95		4.685	
$9S_{21}$		8283.85		11.241	I	$1T_{56}$	8631.00 ± 8.00	8634.93		7.081	50
$8S_{23}$	8282.21 ± 8.00	8287.05		4.205	—	$21S_5$		8639.18		4.429	
$1T_{53}$	8290.00 ± 7.00	8292.35		7.023	50	$20S_6$		8642.79		4.644	
$18S_7$		8301.04		1.444		$0S_{83}$	8679.00 ± 8.00	8649.99		8.275	S
$7T_{16}$		8316.07		4.700		$3S_{40}$		8653.08		2.360	M
$6T_{23}$		8324.95		4.358		$25S_1$	8658.00 ± 2.00	8655.17	$1.35 \pm .25$	1.185	ss
$1S_{53}$		8329.31		7.545		$8T_6$		8656.08		4.964	
$7S_{27}$	8342.30 ± 1.70	8344.50		3.892	W	$6T_{25}$		8656.69		4.309	
$4S_{38}$	8342.15 ± 2.20	8346.24		4.242	W	$9S_{22}$		8657.36		11.248	I
$2S_{44}$	8360.54 ± 4.30	8369.08		6.124	W	$2S_{46}$	8656.02 ± 4.80	8661.61		6.173	W
$0S_{80}$	8395.00 ± 8.00	8370.50		8.208	S	$4S_{40}$	8663.48 ± 2.80	8662.79		4.335	W
$21S_4$		8373.62		11.544	I	$2T_{44}$	8654.00 ± 8.00	8663.95		5.200	50
$2T_{42}$	8370.00 ± 8.00	8382.36		5.108	WT	$19S_7$		8671.41		10.411	I
$5T_{27}$		8382.56		4.183		$10S_{21}$	8675.90 ± 2.50	8673.48		2.922	W
$0T_{71}$	8392.00 ± 4.00	8383.00		7.516	T	$1S_{56}$		8678.47		7.713	

TABLE 4.1 continued

Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}^a	q_{PREM}	ref. ^b	Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}	q_{PREM}	ref.
$_{12}S_{16}$	8689.38 ± 1.80	8691.79		2.225	W	$8T_{13}$		9017.75		4.900	
$8T_7$		8692.62		4.959		$1S_{59}$		9022.73		7.864	
$_{20}S_7$		8706.24		5.731		$_{25}S_2$	9026.00 ± 2.00	9022.91	$1.21 \pm .10$	1.269	WT
$0T_{74}$	8723.00 ± 5.00	8712.37		7.489	T	$0S_{87}$	9057.00 ± 9.00	9023.43		8.350	S
$3T_{38}$	8702.00 ± 8.00	8715.17		4.622	50	$3T_{40}$	9020.00 ± 10.0	9026.32		4.653	WT
$4T_{33}$	8711.00 ± 8.00	8721.15		4.395	50	$9S_{23}$		9030.75		11.255	I
$7S_{29}$	8712.29 ± 3.00	8722.02		3.870	W	$_{10}S_{23}$	9025.00 ± 10.0	9038.42		3.756	WT
$_{11}S_{20}$	8724.19 ± 5.00	8726.69		3.927	W	$0T_{77}$	9054.00 ± 5.0	9041.69	$8.41 \pm .35$	7.461	T,G
$5S_{36}$	8726.95 ± 3.50	8732.50		4.237	W	$_{19}S_9$	9044.00 ± 8.00	9047.16		1.632	WT
$8T_8$		8734.26		4.953		$5S_{38}$	9043.67 ± 2.70	9048.53		4.258	W
$_{14}S_{13}$	8731.00 ± 3.50	8734.78		2.097	WT	$4T_{35}$	9050.00 ± 7.00	9052.02		4.438	WT
$_{16}S_{11}$	8729.00 ± 5.00	8736.47		1.816	WT	$_{10}S_0$	9059.00 ± 2.00	9056.12		1.206	
$7T_{30}$		8740.51		4.677		$_{13}S_{17}$	9053.82 ± 3.00	9063.12		2.083	W
$0S_{84}$	8774.00 ± 8.00	8743.27	$9.20 \pm .30$	8.296	S,O	$2T_{47}$	9071.00 ± 8.00	9078.56		5.359	50
$_{18}S_9$	8735.00 ± 7.00	8747.40		4.299	WT	$4S_{42}$		9085.50		2.346	M
$1T_{57}$	8748.00 ± 10.0	8748.74		7.097	50	$8S_{28}$	9074.00 ± 7.00	9086.85		4.143	—
$_{15}S_{13}$		8749.54		11.679	I	$1T_{60}$	9087.00 ± 9.00	9089.22		7.137	50
$5T_{29}$		8751.14		4.257		$8T_{14}$		9089.28		4.882	
$_{13}S_{16}$	8741.38 ± 1.60	8752.25		2.313	W	$2S_{49}$		9094.25		6.239	WT
$8S_{26}$	8753.56 ± 7.00	8756.06		4.191	—	$7S_{31}$	9089.29 ± 2.80	9095.12		3.875	W
$6S_{32}$	8759.31 ± 3.20	8761.80		3.991	W	$_{16}S_{13}$	9088.37 ± 5.50	9095.71		3.627	
$8T_9$		8780.96		4.946		$6S_{34}$	9092.08 ± 3.00	9097.31		4.004	W
$1S_{57}$		8793.72		7.765		$5T_{31}$	9096.00 ± 7.00	9105.98		4.337	
$_{19}S_8$	8792.95 ± 8.00	8797.85		3.536		$_{11}S_{22}$	9103.06 ± 2.40	9111.59		3.153	W
$2T_{45}$	8785.00 ± 6.00	8803.22		5.250	WT	$7T_{23}$		9111.72		4.629	
$2S_{47}$	8807.27 ± 5.10	8806.64		6.196	W	$0S_{88}$	9151.00 ± 9.00	9116.92		8.365	S
$_{17}S_{10}$		8817.00		11.146	I	$3S_{43}$	9138.26 ± 2.30	9131.62		4.510	W
$3S_{41}$	8823.12 ± 2.60	8819.77		4.400	W	$1S_{60}$		9136.57		7.911	
$0T_{75}$	8834.00 ± 5.00	8822.15		7.480	T	$_{12}S_{18}$	9138.54 ± 5.00	9145.99		3.073	
$6T_{26}$		8831.46		4.290		$_{17}S_{12}$	9145.50 ± 3.00	9151.28		2.163	WT
$8T_{10}$		8832.70		4.937		$0T_{78}$	9164.00 ± 6.00	9151.46		7.451	T
$_{18}S_{10}$		8836.23		4.985		$8T_{15}$		9165.71		4.862	
$0S_{85}$	8867.00 ± 8.00	8836.60		8.315	S	$_{14}S_{15}$	9163.00 ± 4.00	9168.47		3.598	
$_{21}S_6$	8849.50 ± 0.80	8850.77	$1.58 \pm .10$	1.351	WT	$_{21}S_7$	9170.05 ± 1.50	9173.79	$1.58 \pm .10$	1.250	WT
$7T_{21}$		8859.08		4.665		$3T_{41}$	9166.00 ± 8.00	9180.76		4.670	WT
$1T_{58}$	8857.00 ± 9.00	8862.38		7.112	50	$6T_{28}$		9194.29		4.271	
$_{10}S_{22}$	8864.67 ± 2.50	8868.46		3.417	W	$9S_{24}$	9190.00 ± 5.00	9196.93		3.929	WT
$4S_{41}$		8869.34		2.348	M	$1T_{61}$	9196.00 ± 8.00	9202.44		7.148	50
$3T_{39}$	8858.00 ± 8.00	8871.12		4.637	WT	$5S_{39}$	9200.63 ± 3.20	9206.92		4.259	W
$4T_{34}$	8873.00 ± 6.00	8887.20		4.416	WT	$0S_{89}$	9246.00 ± 9.00	9210.47		8.380	S
$8T_{11}$		8889.44		4.927		$2T_{48}$	9205.00 ± 8.00	9214.61		5.416	50
$5S_{37}$	8884.50 ± 2.90	8890.43		4.250	W	$4T_{36}$	9208.00 ± 8.00	9215.65		4.462	50
$_{11}S_{21}$	8896.96 ± 3.35	8904.37		3.606	W	$_{24}S_3$		9218.64		10.287	I
$1S_{58}$		8908.46		7.816		$2S_{50}$		9236.83		6.261	
$7S_{30}$	8902.52 ± 2.40	8909.23		3.870	W	$7T_{24}$		9246.21		4.601	
$8S_{27}$	8912.00 ± 7.50	8919.60		4.170	WT	$8T_{16}$		9247.01		4.839	
$6S_{33}$	8926.92 ± 2.70	8929.44		4.000	W	$1S_{61}$		9250.01		7.955	
$0S_{86}$	8962.00 ± 8.00	8929.99	$8.90 \pm .30$	8.333	S,G	$8S_{29}$	9242.65 ± 8.00	9257.65		4.114	—
$5T_{30}$		8930.17		4.298		$0T_{79}$	9274.00 ± 6.00	9261.21		7.441	T
$0T_{76}$	8943.00 ± 5.00	8931.93		7.471	T	$_{20}S_8$		9262.17		11.298	I
$_{12}S_{17}$	8930.60 ± 5.00	8933.91		2.612	WT	$6S_{35}$	9257.99 ± 3.70	9265.32		4.006	W
$2T_{46}$	8932.00 ± 7.00	8941.43		5.303	WT	$_{15}S_{14}$		9270.16		11.706	I
$_{23}S_4$	8937.00 ± 2.00	8941.57	$1.35 \pm .10$	1.235	WT	$5T_{32}$		9279.00		4.372	
$_{17}S_{11}$		8942.70		3.946		$7S_{32}$	9279.15 ± 2.30	9279.63		3.882	W
$_{16}S_{12}$		8944.33		3.522		$3S_{44}$	9290.13 ± 2.50	9286.52		4.564	W
$2S_{48}$		8950.86		6.218	WT	$_{23}S_5$	9290.00 ± 0.80	9289.58	$1.15 \pm .10$	1.112	WT
$8T_{12}$		8951.13		4.914		$4S_{43}$		9301.61		2.345	M
$1T_{59}$	8970.00 ± 8.00	8975.87		7.125	50	$0S_{90}$	9339.00 ± 9.00	9304.07		8.394	S
$3S_{42}$	8976.89 ± 2.45	8976.05		4.455	W	$_{16}S_{14}$		9304.36		2.698	
$7T_{22}$		8982.76		4.650		$1T_{62}$	9306.00 ± 7.00	9315.53		7.157	50
$_{14}S_{14}$	8982.00 ± 4.00	8985.12		3.019	WT	$_{12}S_{19}$	9320.02 ± 7.00	9329.10		3.469	WT
$_{22}S_5$		9005.25		11.708	I	$8T_{17}$		9333.20		4.813	
$6T_{27}$		9011.09		4.277		$3T_{42}$	9323.00 ± 8.00	9334.44		4.688	WT

TABLE 4.1 continued

Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}^a	q_{PREM}	ref. ^b	Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}	q_{PREM}	ref.
$_{14}S_{16}$	9334.00 ± 4.00	9337.18		3.543		$_{9}T_4$		9621.48		4.773	
$_{11}S_{23}$	9327.00 ± 2.00	9341.10		2.863	W	$_{3}T_{44}$	9615.00 ± 7.00	9639.58		4.723	WT
$_{2}T_{49}$	9338.00 ± 8.00	9349.55		5.475	50	$_{7}S_{34}$	9636.83 ± 2.70	9644.54		3.899	W
$_{9}S_{25}$	9343.00 ± 6.00	9351.55		4.017		$_{9}T_5$		9645.80		4.762	
$_{19}S_{10}$	9353.00 ± 2.00	9357.39	$1.98 \pm .25$	1.479	WT	$_{23}S_6$		9646.41		7.094	I
$_{1}S_{62}$		9363.08		7.997		$_{12}S_{21}$	9642.81 ± 8.00	9647.81		3.867	WT
$_{26}S_1$		9364.82		11.554	I	$_{24}S_5$		9647.92		4.675	
$_{5}S_{40}$	9360.49 ± 3.90	9365.67		4.256	W	$_{19}S_{11}$	9644.00 ± 4.00	9653.75		1.872	
$_{0}T_{80}$	9384.00 ± 6.00	9370.95		7.430	T	$_{1}T_{65}$	9647.00 ± 7.00	9654.15		7.180	50
$_{13}S_{18}$	9365.05 ± 1.40	9371.79		2.038	W	$_{2}S_{53}$		9659.62		6.326	
$_{4}T_{37}$	9390.00 ± 7.00	9378.12		4.485	WT	$_{9}S_{27}$	9657.00 ± 7.00	9660.38		4.088	
$_{2}S_{51}$		9378.59		6.282		$_{13}S_{19}$	9663.55 ± 1.20	9671.80		2.054	W
$_{6}T_{29}$		9379.50		4.272		$_{9}T_6$		9674.96		4.750	
$_{18}S_{11}$		9384.31		11.660	I	$_{0}S_{94}$	9718.00 ± 9.00	9678.99		8.438	S
$_{7}T_{25}$		9386.54		4.564		$_{5}S_{42}$	9679.32 ± 3.10	9684.42		4.239	W
$_{0}S_{91}$	9435.00 ± 9.00	9397.73		8.407	S	$_{7}T_{27}$		9686.26		4.465	
$_{10}S_{24}$		9404.05		11.263	I	$_{4}T_{39}$	9679.00 ± 9.00	9699.82		4.530	WT
$_{8}T_{18}$		9424.29		4.783		$_{0}T_{83}$	9715.00 ± 7.00	9700.14		7.398	T
$_{25}S_3$		9424.61		2.646		$_{24}S_6$		9700.17		4.089	
$_{1}T_{63}$	9422.00 ± 7.00	9428.51		7.166	50	$_{1}S_{65}$		9700.36		8.113	
$_{8}S_{30}$	9421.74 ± 8.00	9431.74		4.083	—	$_{9}T_7$		9708.96		4.735	
$_{6}S_{36}$	9423.84 ± 3.10	9433.35		4.005	W	$_{17}S_{14}$	9705.00 ± 5.00	9709.08		2.165	WT
$_{17}S_{13}$	9429.00 ± 3.00	9435.93		1.804	WT	$_{22}S_7$		9716.00		4.489	
$_{3}S_{45}$	9441.41 ± 3.00	9440.80		4.618	W	$_{8}T_{21}$		9727.24		4.671	
$_{5}T_{33}$	9440.00 ± 7.00	9449.69		4.402		$_{4}S_{45}$		9733.71		2.343	M
$_{7}S_{33}$	9457.46 ± 2.10	9462.76		3.890	W	$_{3}S_{47}$	9750.64 ± 2.50	9747.60		4.724	W
$_{1}S_{63}$		9475.81		8.038		$_{9}T_8$		9747.78		4.719	
$_{0}T_{81}$	9495.00 ± 6.00	9480.69		7.420	T	$_{2}T_{52}$	9740.00 ± 10.0	9747.83		5.661	50
$_{2}T_{30}$	9479.00 ± 10.0	9483.40		5.536	50	$_{14}S_{18}$	9741.70 ± 5.50	9748.85		2.707	WT
$_{27}S_1$	9494.00 ± 3.00	9485.84	$1.35 \pm .24$	1.542	WT	$_{25}S_4$		9749.46		3.347	
$_{3}T_{43}$	9472.00 ± 7.00	9487.38		4.706	WT	$_{6}T_{31}$		9750.03		4.287	
$_{0}S_{92}$	9529.00 ± 9.00	9491.43		8.418	S	$_{1}T_{66}$	9753.70 ± 9.00	9766.83		7.187	50
$_{12}S_{20}$	9487.18 ± 7.00	9493.18		3.724	WT	$_{22}S_8$		9766.99		4.488	
$_{21}S_8$	9492.00 ± 3.00	9496.96	$1.45 \pm .10$	1.497	WT	$_{6}S_{38}$	9760.55 ± 3.00	9768.94		4.004	W
$_{9}S_{26}$	9503.00 ± 5.00	9505.55		4.063	WT	$_{0}S_{95}$	9813.00 ± 9.00	9772.85	$9.45 \pm .30$	8.447	S,O
$_{4}S_{44}$		9517.68		2.344	M	$_{10}S_{25}$		9777.27		11.271	I
$_{2}S_{52}$		9519.52		6.304	WT	$_{5}T_{35}$		9785.86		4.443	
$_{8}T_{19}$		9520.30		4.749		$_{16}S_{15}$		9786.62		11.714	I
$_{5}S_{41}$	9511.00 ± 4.00	9524.84		4.249	W	$_{8}S_{32}$	9767.23 ± 7.00	9788.23		4.023	—
$_{14}S_{17}$	9514.00 ± 4.50	9526.65		3.162		$_{20}S_9$	9810.00 ± 10.0	9790.64		3.485	WT
$_{7}T_{26}$		9533.08		4.519		$_{3}T_{45}$	9776.00 ± 9.00	9791.08		4.741	50
$_{4}T_{38}$	9523.00 ± 7.00	9539.49		4.509	50	$_{9}T_9$		9791.40		4.700	
$_{1}T_{64}$	9533.00 ± 8.00	9541.38		7.174	50	$_{2}S_{54}$		9798.87		6.348	WT
$_{6}T_{30}$		9565.17		4.279		$_{12}S_{22}$		9799.02		3.939	
$_{9}T_1$		9577.68		4.792		$_{0}T_{84}$	9825.00 ± 7.00	9809.85		7.387	T
$_{11}S_{24}$	9572.88 ± 2.00	9578.38		2.757	W	$_{1}S_{66}$		9812.22		8.148	
$_{28}S_1$		9579.07		4.744		$_{11}S_{25}$	9811.00 ± 5.00	9814.25		2.769	WT
$_{0}S_{93}$	9624.00 ± 9.00	9585.19		8.429	S	$_{9}S_{28}$	9814.28 ± 8.50	9816.78		4.100	
$_{26}S_2$		9586.48		4.754		$_{7}S_{35}$	9820.29 ± 2.70	9824.99		3.908	W
$_{9}T_2$		9587.42		4.788		$_{21}S_9$		9828.72		4.309	
$_{1}S_{64}$		9588.23		8.077		$_{8}T_{22}$		9838.26		4.626	
$_{0}T_{82}$	9605.00 ± 6.00	9590.42		7.409	T	$_{9}T_{10}$		9839.82		4.680	
$_{3}S_{46}$	9603.01 ± 2.70	9594.48		4.672	W	$_{5}S_{43}$	9835.77 ± 5.00	9844.43		4.227	W
$_{15}S_{15}$	9595.53 ± 2.00	9597.78		2.004	W	$_{7}T_{28}$		9846.40		4.402	
$_{6}S_{37}$	9599.00 ± 3.50	9601.27		4.005	W	$_{4}T_{40}$	9844.20 ± 8.00	9859.20		4.550	50
$_{9}T_3$		9602.02		4.781		$_{27}S_2$	9871.00 ± 3.00	9865.33	$1.10 \pm .12$	1.266	WT
$_{22}S_6$		9603.76		6.315		$_{0}S_{96}$	9908.00 ± 9.00	9866.75		8.455	S
$_{26}S_3$		9604.98		4.727		$_{22}S_9$		9871.01		9.425	I
$_{8}S_{31}$	9593.74 ± 8.00	9608.74		4.052	—	$_{2}T_{53}$	9870.00 ± 12.0	9878.43		5.725	50
$_{24}S_4$		9615.50		4.537		$_{1}T_{67}$	9869.20 ± 7.00	9879.42		7.192	50
$_{2}T_{31}$	9610.00 ± 10.0	9616.16		5.598	50	$_{11}S_0$	9892.00 ± 5.00	9887.92		1.202	
$_{5}T_{34}$		9618.51		4.426		$_{9}T_{11}$		9893.01		4.658	
$_{8}T_{20}$		9621.27		4.712		$_{18}S_{12}$		9894.79		2.975	

TABLE 4.1 continued

Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}^a	q_{PREM}	ref. ^b	Mode	f_{obs} μHz	f_{PREM} μHz	q_{obs}	q_{PREM}	ref.
$_{20}S_{10}$		9900.15		4.259		$_{4}S_{46}$		9949.71		2.343	M
$_{3}S_{48}$	9908.12 ± 4.00	9900.17		4.775	W	$_{12}S_{23}$		9950.43		3.965	
$_{26}S_{4}$		9910.84		9.693	I	$_{9}T_{12}$		9950.95		4.634	
$_{0}T_{85}$	9935.00 ± 8.00	9919.55		7.376	T	$_{5}T_{36}$	9943.00 ± 7.00	9952.11		4.453	
$_{1}S_{67}$		9923.84		8.182		$_{8}T_{23}$		9954.41		4.579	
$_{15}S_{16}$	9921.25 ± 1.50	9926.89		1.857	W	$_{0}S_{97}$	10003.0 ± 9.00	9960.70		8.461	S
$_{6}T_{32}$		9933.27		4.293		$_{13}S_{20}$	9953.63 ± 1.50	9961.02		2.114	W
$_{6}S_{39}$	9928.85 ± 2.30	9936.22		4.004	W	$_{8}S_{33}$		9969.78		3.997	
$_{2}S_{55}$		9937.26		6.372		$_{9}S_{29}$		9975.13		4.106	
$_{17}S_{15}$	9931.93 ± 5.50	9938.07		2.835	W	$_{23}S_{7}$		9983.94		1.497	
$_{3}T_{46}$	9926.00 ± 7.00	9941.90		4.758	WT	$_{1}T_{68}$		9991.93		7.197	
$_{19}S_{12}$		9945.85		11.627	I	$_{14}S_{19}$	9979.63 ± 5.00	9994.61		2.461	

a) $q \equiv 1000/Q$; b) Reference number or mode type; c) Coriolis or second order rotation correction applied.

G Gabi Laske – personal communication (surface wave analysis)

H Dalton et al, 2006 (surface wave analysis)

O Selby and Woodhouse 2000 (surface wave analysis)

RR Resovsky and Ritzwoller (web page)

ss singlet stripping

A improved results from Andaman earthquake (singlet stripping)

W regional multiplet stripping (Widmer-Schmidrig, 2002)

WT multiplet stripping (Widmer Thesis)

tvf q estimated using time variable filtering

T see tab.t.fundamentals on the rem webpage (combination of several studies)

S see tab.t.fundamentals on the rem webpage (combination of several studies)

I denotes a mode whose energy is trapped within the inner core or on the inner-core boundary (unlikely to be observed).

M denotes a Stoneley mode trapped on the core-mantle boundary (unlikely to be observed).